Magnetic Properties of Diabase Dikes

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Abstract. A study has been made of the magnetic properties of several diabase dikes of Precambrian age from the Canadian shield. These dikes are of widely differing ages and belong to several swarms having widely different strike directions. It was found that the direction of remanent magnetization of the various dikes was quite widely scattered before they were demagnetized in a-c fields of 100 to 200 oersteds. However, after demagnetization a weak remanent magnetization was found whose direction was stable in a-c fields up to 500 oe. In almost all the cases studied the direction of this stable magnetization was found to lie close to the plane of the dike, although the various dike swarms had different inclinations within the plane. It was shown experimentally that typical samples acquired a large, soft TRM at 585°C when cooled in the earth's field. A second weak but very stable component was also found which had a Curie temperature considerably less than 585°C. It is postulated that the component with the high Curie temperature produced internal demagnetization which tended to cancel out the component of the earth's magnetic field normal to the dike. The component with the lower Curie temperature then acquired a magnetic moment parallel to the internal field of the dike, which would be nearly in the plane of the dike. This stable component may have been preserved since the time of formation of the dike.

Introduction. In the Canadian shield there are several swarms of diabase dikes which have long been known to geologists. However, wide-scale aeromagnetic coverage is revealing the presence of a far greater number of these than had been suspected previously. Moreover, the magnetometer is proving that some of these dikes are as much as 300 miles long, although only a few hundred feet wide. The dikes are all considered to be of Precambrian or very early Cambrian age, but there are known to be wide differences in age. The individual swarms have varying strike directions and have been classified separately as, for example, the Matachewan, Nipissing, and Grenville types. (For a more complete discussion of the dike classification, see *Strangway* [1960].)

In view of the geological importance of these dikes, it was decided that a study of the magnetic properties might be useful in throwing light on their relative ages as well as on the magnetic properties of dikes in general.

Sample collection and measurement. One hundred and eighty-six samples were collected from sites shown in the accompanying generalized map of portions of Ontario and Quebec (Fig. 1). Samples were collected from 19 dikes at 14 different sites. Samples from three of the dikes had remanent magnetizations below the limit of measurement (5×10^{-5} emu/cm^3). At each dike approximately ten samples were taken to test for variation in properties across the width of the dike. Each sample was cut into several 1-inch cubes and the direction and intensity of the remanent magnetization were measured. This was done with a spinner magnetometer [*Hood,* 1956] in which the cubes were spun at 40 rev/sec adjacent to a pickup coil. The reversible susceptibility of each sample was also measured in a 40-cps a-c field with a peak intensity of about 1.0 oe.

Fourteen of the 19 dikes were examined in detail. None showed any regular variation in the value of susceptibility (k), in the value of remanent magnetization (J_r), or in the ratio of the remanent to the induced magnetization ($Q = J_r/kH$) when the values across the width of the dike were compared. This was somewhat surprising in view of the observed large gradation of grain size from the edges to the middle of any one dike. This may be due to the fact (as noted in thin sections and polished sections) that the coarse-grained portions of the dikes

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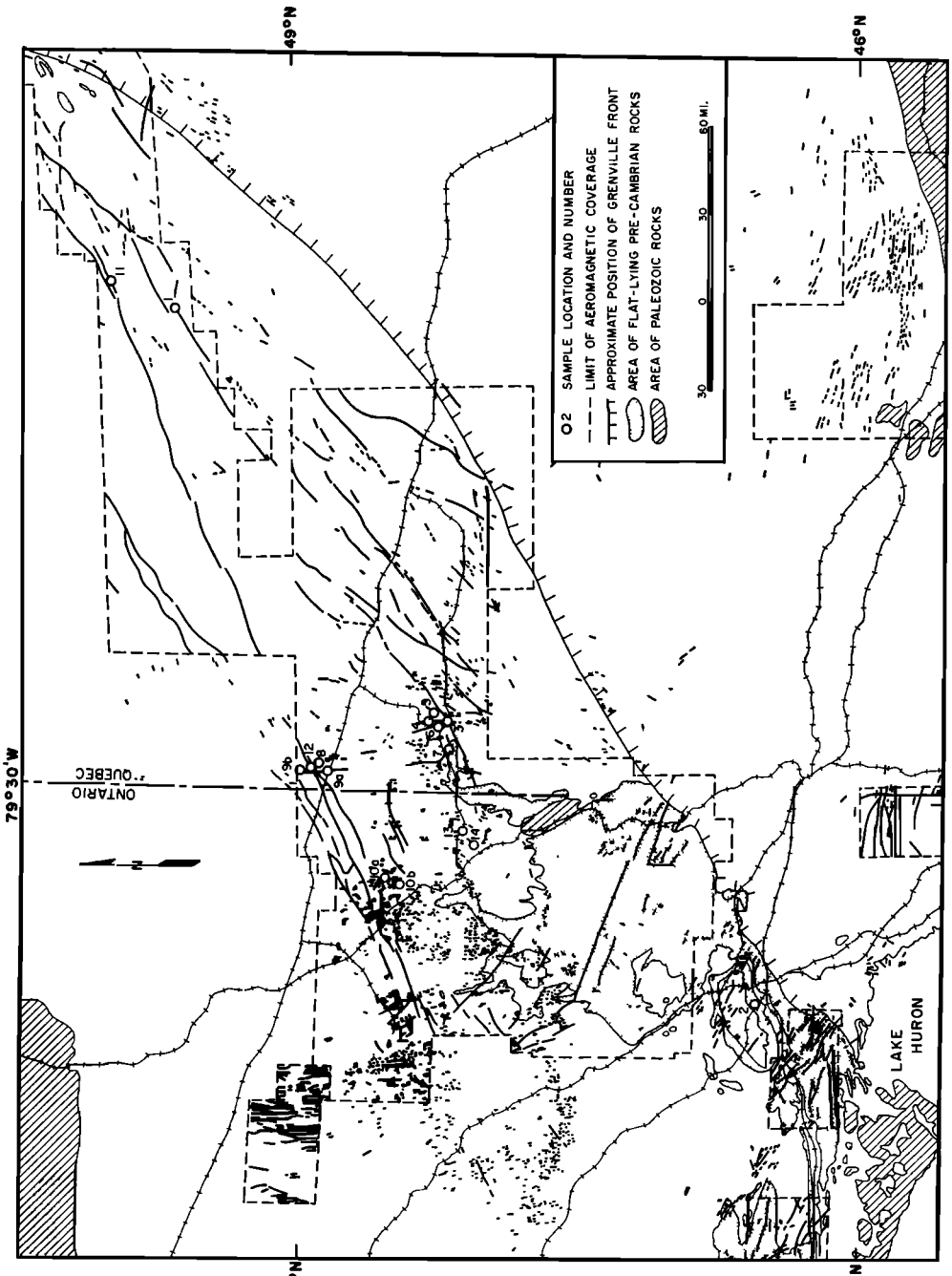


Fig. 1. Younger diabase dikes of N.E. Ontario and N.W. Quebec.

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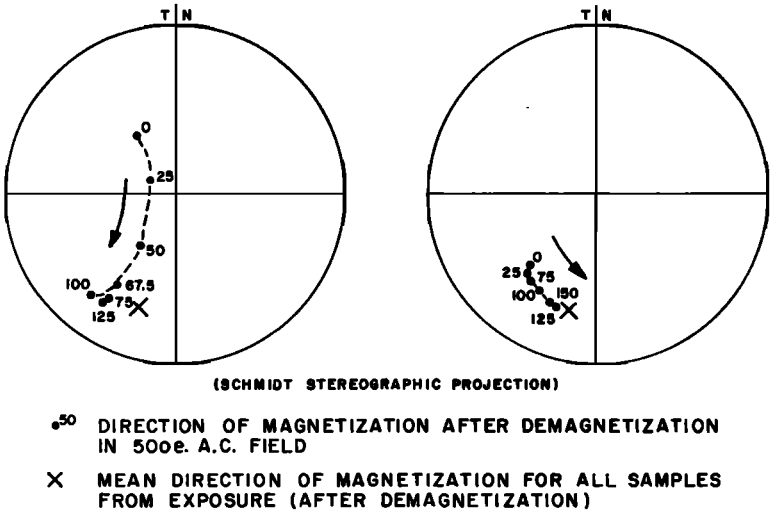


Fig. 2. Change in direction of remanent magnetization with increasing a-c demagnetization field.

contained intimate intergrowths of magnetite and ilmenite while the fine-grained portions contained discrete particles. Consequently, the magnetite and ilmenite particles in the center of the dike had about the same dimensions as the individual grains at the edge of the dike.

Because the initial directions of magnetization were usually found to be quite widely scattered, at least one cube from every sample was subjected to a-c demagnetization. The a-c (60 cps) demagnetization was carried out at various field strengths in a space that was free of d-c field (to within about 5 per cent). The purpose of eliminating most of the earth's field was to avoid the spurious 'anhysteretic' magnetization described by *Rimbert* [1957] and others. The problem of 'anhysteretic' magnetization proved to be very serious because of the large soft components of magnetization that were encountered. It was found that by demagnetizing a second time after rotating the sample through 180°, so that any residual d-c field acted in opposite directions, it was possible to detect the presence of such a spurious magnetization. If after demagnetization the two magnetization vectors were found to differ by less than 60°, a simple averaging process reproduced the true direction of stable magnetization with considerable accuracy. This double demagnetization was repeated along each of the three cube axes and the direction of stable magnetization determined.

For a few samples from each dike this process was carried out using a-c fields of increasing strength. It was found generally that at a-c fields of 100 to 200 oe, the direction of magnetization had become stable to increases in field strength. In two of the dikes no magnetization was detectable after demagnetization in fields over 50 oe, probably indicating that the stable magnetization found in most of the dikes was not present in these two. Figure 2 shows two examples of how the direction of magnetization approached a final value. After the minimum field necessary to reach stable magnetization was determined for any given dike, at least one, and usually two, cubes from every sample was demagnetized using this minimum field strength. A second measurement using a higher field strength was usually carried out to confirm the direction of stable magnetization.

Figure 3 shows data for two exposures from one dike. It can be seen that the grouping of the results is much better after than before demagnetization, which lends support to the idea that a soft component of magnetization having a random direction has been removed by the demagnetization process. This same improvement in grouping after demagnetization was observed in most of the exposures studied.

One of the major unexplained problems in rock magnetism is that of magnetic reversals. In Figure 3D three cubes show a stable magnet-

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ization exactly reversed to that of the bulk of the samples. It is difficult to explain these results except on the basis of some still unknown physico-chemical process. If the explanation were based on a reversal of the magnetic field during cooling, one would expect the reversed samples to be symmetrically placed on the two sides of the dike. However, these samples came from one edge of the dike while the other edge showed only a normal polarization. It is suggested that some unknown physico-chemical process has caused this reversal (and several

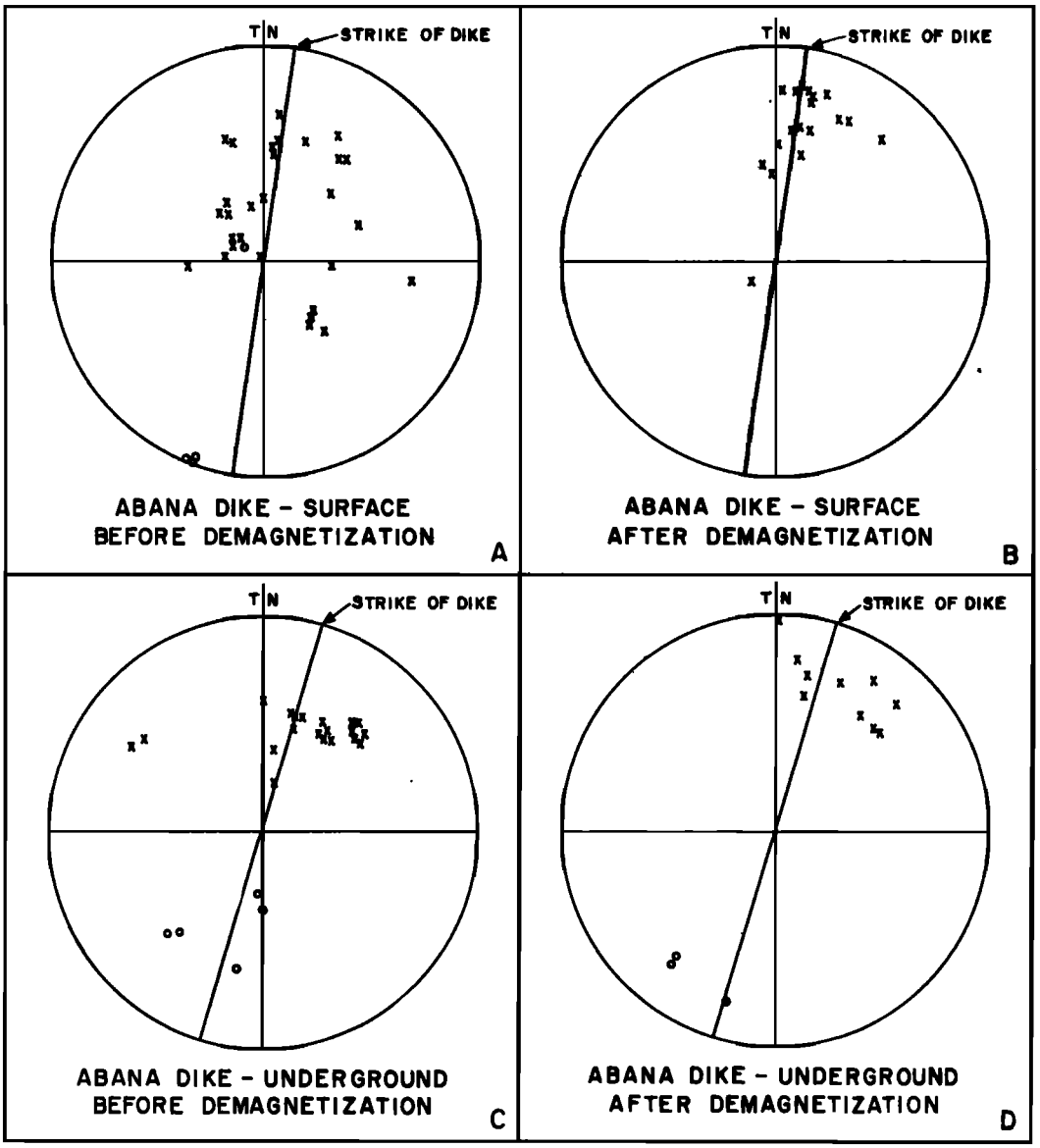


Fig. 3. Stereographic plots of directions of remanent magnetization showing effect of demagnetizing.

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TABLE 1. Summary of Paleomagnetic Results after Demagnetization

Dike No.	Type of Dike	Strike of Dike, deg	Avg. Declination of J_r, deg	Avg. Inclination of J_r, deg	No. of Samples	α, deg	K
1	ENE dike parallel to Grenville Front	258	263	−56	9	10.8	23.5
2	NW Sudbury olivine dike	297	266	−8	8	25.0	5.1
3	N-S dike (Noranda area)	256	255	−61	7	15.9	12.8
4	N-S dike (Noranda area) (Matachewan type)	191	215	+8	8	25.9	4.9
5	E-W dike (Noranda area)	226	246	−47	5	18.0	12.9
6	N-S dike (Noranda area)	164	161	+19	8	16.1	12.5
7	N-W dike (Noranda area)	154	190	+10	5	40.5	1.2
8	ENE dike parallel to Grenville Front	251	266	−42	13	8.8	24.5
9 (2 exposures)	N-S dike (Normetal area) (Abana dike)	16	18	−30	31	5.3	24.2
10 (4 dikes)	N-S Matachewan dikes	176 & 197	186	+3	21	19.2	6.5
11	ENE dike parallel to Grenville Front	243	233	−69	1		
12, 13, 14	(Remanent magnetic intensities less than 5×10^{-5} emu/cm^3)						

similar ones found in this study), and the present calculations have been made assuming all samples to be normally magnetized.

Results. When the direction of stable remanent magnetization of each sample was plotted, it was observed that the mean direction of magnetization was always close to the plane of the dike involved. The inclination within the plane varied from one dike swarm to another, but there seemed to be a strong correlation with the plane in each case. This is shown in Figures 3 and 4 and Table 1. Those specimens which had extremely high Q values were rejected, as they were undoubtedly remagnetized by lightning strokes. However, the rest of the samples which had been demagnetized were analyzed statistically using the *Fisher* [1953] analysis for dispersion on a sphere. As is commonly done in rock magnetism, a 5 per cent limit of confidence was placed on the results so that there is a 95 per cent chance that the true mean lies within α degrees of the calculated mean. Table 1 includes the results of these computations for the dikes sampled and also gives the strike direction and the location from which the samples were obtained. The value K is a precision parameter which gives an indication of the dispersion of the directions of magnetization. When $K = 1$, the directions are completely randomly distributed, and as K becomes larger the grouping improves.

With the exception of dikes 2 and 7, the directions of the average declinations lie within 24°, and in most cases within 15°, of the strike directions of the respective dikes. In the case of dikes 2 and 7, no stable component of magnetization was found, since the intensity of magnetization became extremely small and unmeasurable before a stable direction could be reached. As will be pointed out later, it is possible that the applied field may have been nearly normal to the plane of these dikes at the time they were last cooled through their Curie temperatures. It is interesting to note that dikes 2 and 7 both trend toward the northwest, and they apparently bear no relation to any of the other dike swarms studied.

The relation between strike direction and direction of remanent magnetization has been summarized on an equal-area stereographic contour plot. In Figure 4 the results of 132 demagnetization measurements have been compiled with the strike of the respective dikes as the reference plane. Contouring is carried out in the usual fashion (1 per cent of the total area of the projection contains a given percentage of the total number of points present). In this plot it is evident that up to 12 per cent of the total number of points are contained within 1 per cent of the area. The correlation of the remanent magnetization vector with the strike of the dike is very obvious. No contour

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intervals are located more than about 20° from the plane defined by the dikes.

There have been a few other studies of similar dikes with a range of ages and in various parts of the world. Among these studies are those by *Graham* [1953] on dikes in northern Michigan, by *Bruckshaw and Robertson* [1948] on dikes in England and Scotland, and by *Gough* [1956] on dikes in South Africa. The samples in most of these studies have not been subjected to routine demagnetization, but fairly consistent results have nevertheless been found. In the present study it was found that when the initial results were well grouped, the demagnetization of the samples did not change the mean direction a great deal. All the results obtained in the present study and all those by the above authors have been compiled in Figure 5 on an equal-area contour map. Once again, there is a very definite correlation between the direction of stable magnetization and the strike of the respective dike. The concentration of points at the center of the projection is the result of the large number of samples collected by Gough of a few dikes of the same age.

Probable physical explanation. The evidence presented here suggests that the most stable component of magnetization in diabase dikes lies close to the plane of the dike. Much work in the field of paleomagnetism assumes *a priori* that the direction of magnetization acquired and retained throughout the lifetime of a sample is parallel to the earth's field at the time of magnetization. However, it seems most unlikely that the direction of the external applied field would always happen to lie parallel to diabase dikes of several different ages and with wide geographic distribution.

Several possible explanations of this anisotropy have been considered. It seems unlikely that there is any preferred grain orientation and hence any possible magnetic anistropy in the dikes. The dikes are all gradational from edge to middle with fine-grained material at the edge of the dike becoming coarse-grained in the middle. This is consistent with a cooling process during which crystallization took place after emplacement. No flow structures have been observed. Thin sections indicate that a random directional distribution of grains exists both at the edge and in the center of the dikes.

Pressure effects cannot be eliminated as a possible explanation of the results, but such causes seem unlikely. If pressure effects were responsible for the magnetic orientation, it is probable that all samples would be magnetized in the same direction (either horizontally or vertically) which is contrary to the large range of inclinations found in this study.

The most plausible explanation appears to

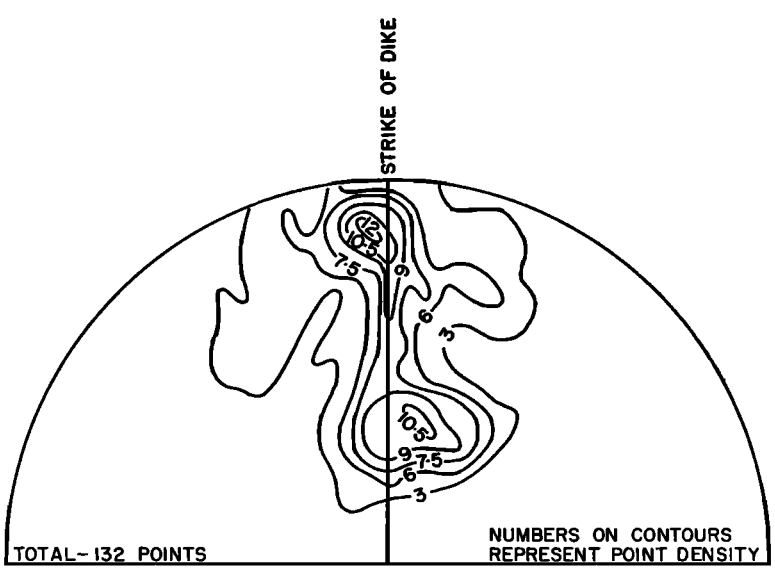


Fig. 4. Stereographic plot of directions of remanent magnetism in dikes from N.E. Ontario and N.W. Quebec showing relationship to strike of the dike.

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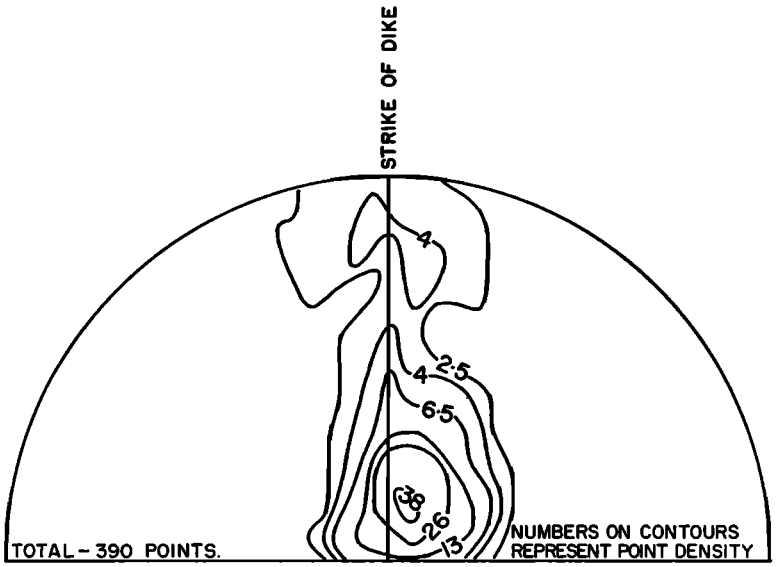


Fig. 5. Stereographic plot of directions of remanent magnetism in dikes for which data have been published showing relationship to strike of the dike.

be one based on demagnetization within an infinite sheet. In Figure 6 we see the various fields that act on any particular point within a dike. First, there is the external field of the earth, H_{ext}. Within a magnetic sheet there is in addition a demagnetizing field H_{dem}, a Lorentz field H_{Lor}, and a field due to the adjacent particles. The demagnetizing field is due to the shape of the body and it is normal to the infinite sheet. This field opposes the external field and so tends to cancel the normal component of the external field. The Lorentz field is a magnetizing field parallel to the magnetization. This field is taken to be that set up within a spherical cavity with the point in consideration at the center of the sphere. The size of this cavity is taken to be sufficiently large that individual magnetic grains outside the cavity exert a field equivalent to that of a uniform magnetization. Finally, it is necessary to add the effect of the individual grains within the Lorentz sphere. If these are randomly distributed, the most probable field within the sphere due to these grains is zero.

The net internal field H_{int}, then, is

$$H_{int} = H_{ext} - H_{dem} + H_{Lor}$$

In terms of components H_{par} parallel and H_{perp} normal to the dike, we have

$$H_{par} = H_{ext} \cos\alpha + \tfrac{4}{3}\pi M \cos\alpha$$

$$H_{perp} = H_{ext} \sin\alpha + \tfrac{4}{3}\pi M \sin\alpha - 4\pi M \sin\alpha$$

where M is the total magnetization $= (kH_{int} + J_r)$ and k = vol. susceptibility.

Since the susceptibility magnetization is dependent on the internal field, this latter magnetization is analogous with a refraction process. From Figure 7 it is apparent that the susceptibility of the diabase dikes does not change the direction of the internal field by more than a few degrees. However, in the same figure a cal-

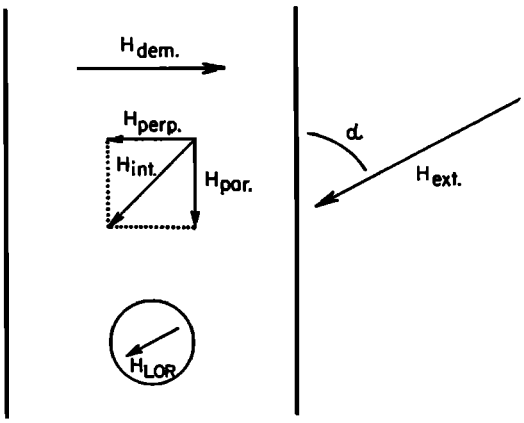


Fig. 6. Fields acting within a magnetic sheet.

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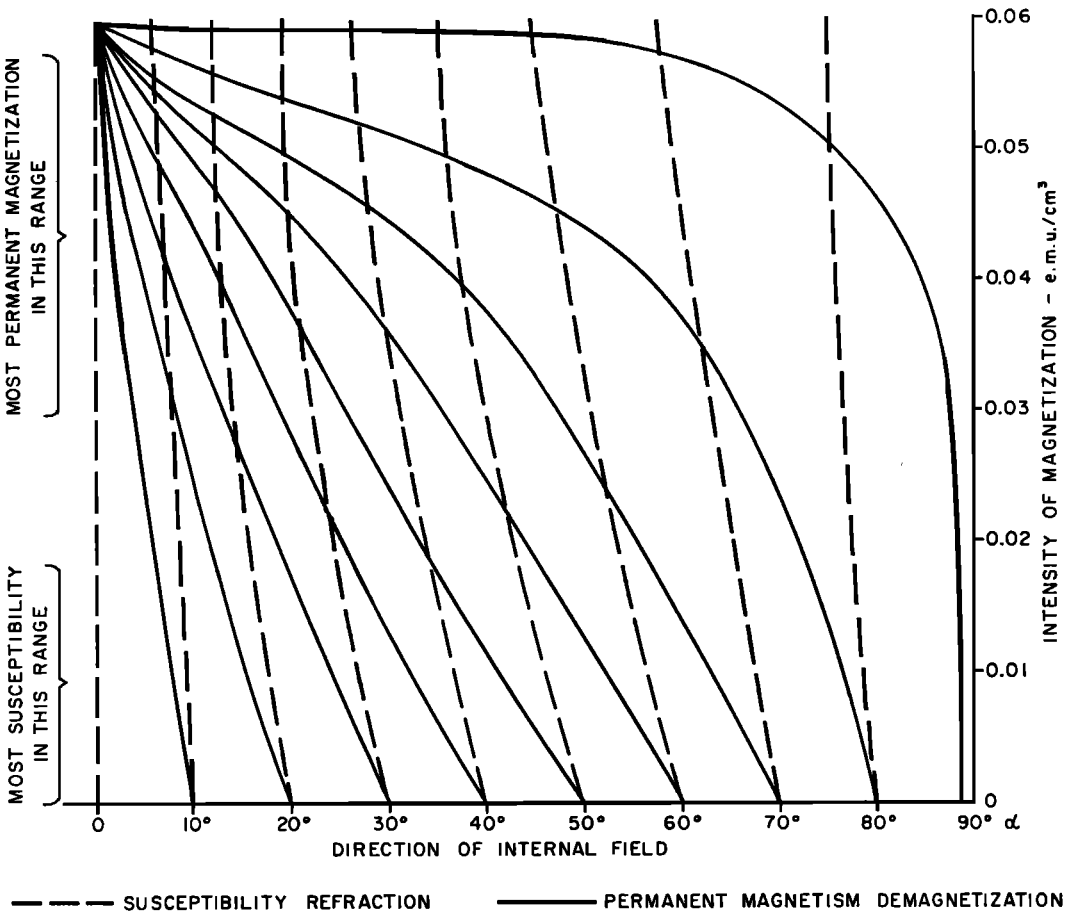


Fig. 7. Variation of direction of internal field with various directions of applied field (α) and various intensities of susceptibility and permanent magnetization.

culation of the direction of the internal field, account having been taken of the demagnetizing and Lorentz fields, shows that the direction of the internal field can be markedly different from that of the external field if a sufficiently large permanent magnetization is present.

If this internal demagnetization is to explain the observed experimental results, a number of conditions must be satisfied:

1. The intensity of freshly acquired remanent magnetism must be relatively large (0.02 to 0.05 emu/cm³).

2. The main body of magnetization must be acquired at a single Curie temperature and be parallel to the applied field, H_{ext}.

3. This magnetization must be sufficiently stable to remain in a fixed direction as the body cools; otherwise the demagnetization effect would be slight. However, this magnetization must be unstable over time intervals of the order of the age of the rock to explain the scattered results found today.

4. Finally, a second more stable component with a lower Curie point must also be present.

As a check on whether these conditions are satisfied, laboratory experiments were carried out on 20 representative samples from the edges and centers of the dikes. The samples were heated above the highest Curie temperature and allowed to cool in the present earth's field. The intensities of remanent magnetization in the cooled samples varied from about 0.015 to 0.058 emu/cm³, with most of the values falling at about 0.03 to 0.04 emu/cm³. The intensity of magnetization after heating was generally increased by a factor of 7 or more, indicating

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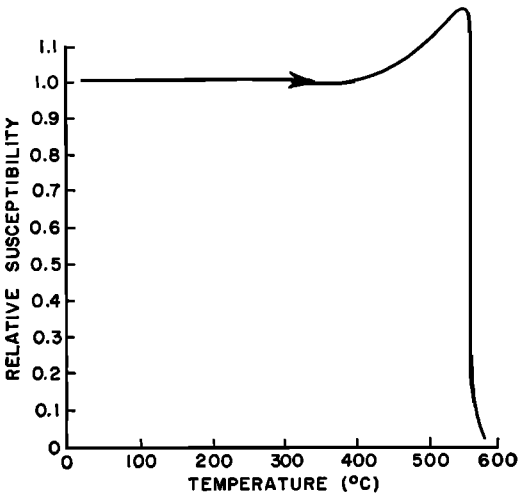


Fig. 8. Typical plot of susceptibility versus temperature (readings taken at approximately 7°C intervals).

that a great deal of the original TRM has decayed in geologic time (or that the earth's field was 7 to 10 times less than it is at present). This intensity of magnetization would certainly be sufficient to reduce greatly the normal component of the internal field of the dike.

While heating these samples, measurements of susceptibility were made at intervals of about 7°C, and almost without exception the curves were typical of a single magnetic component with a Curie temperature of about 580°C (see example in Fig. 8). This suggested that the main component of magnetization was due to magnetite. While most of these samples were being cooled in the earth's field they were rotated at about 5°C below this Curie temperature; therefore their orientation with respect to the applied earth's field was different for the remainder of the cooling process. The direction of magnetization of the samples immediately after cooling was invariably found to be parallel to the original field direction, indicating that the major component of magnetization was determined by the direction of the applied field at the Curie temperature. In other words, the net magnetization of dikes just after cooling is parallel to the external field and is sufficiently stable not to change direction during the cooling process. The main portion of the magnetization is therefore available to set up internal demagnetization.

Following the measurement of the magnitude and direction of the remanent magnetization produced by the cooling process, the samples were subjected to successive a-c demagnetization. It was found that the intensity of magnetization dropped off rapidly, and the direc-

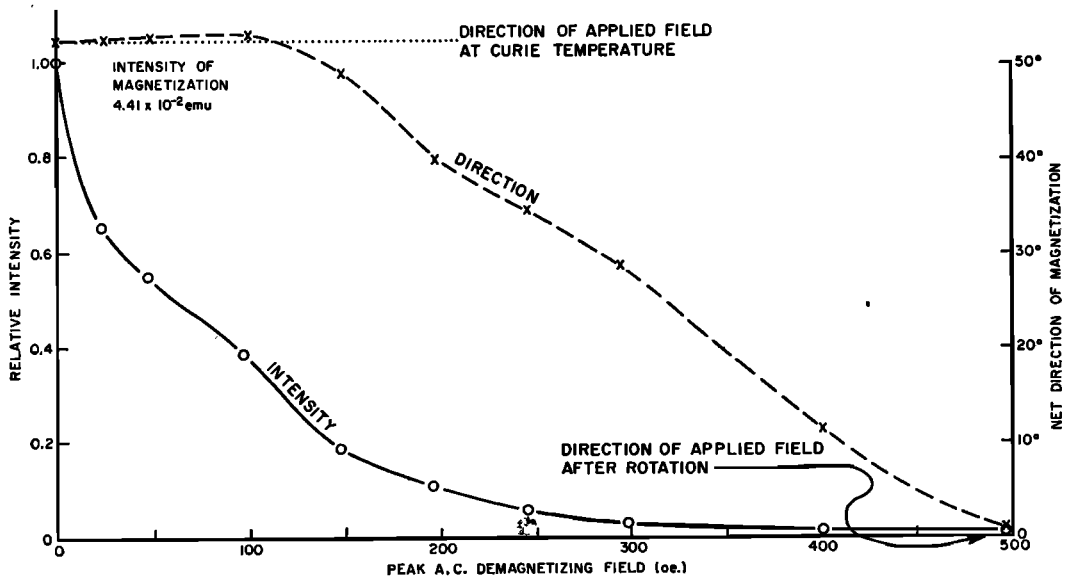


Fig. 9. Results of successive a-c demagnetization of a sample which has been cooled through the Curie temperature in a 0.5-oe field and rotated through a 50° angle at 5°C below the Curie temperature.

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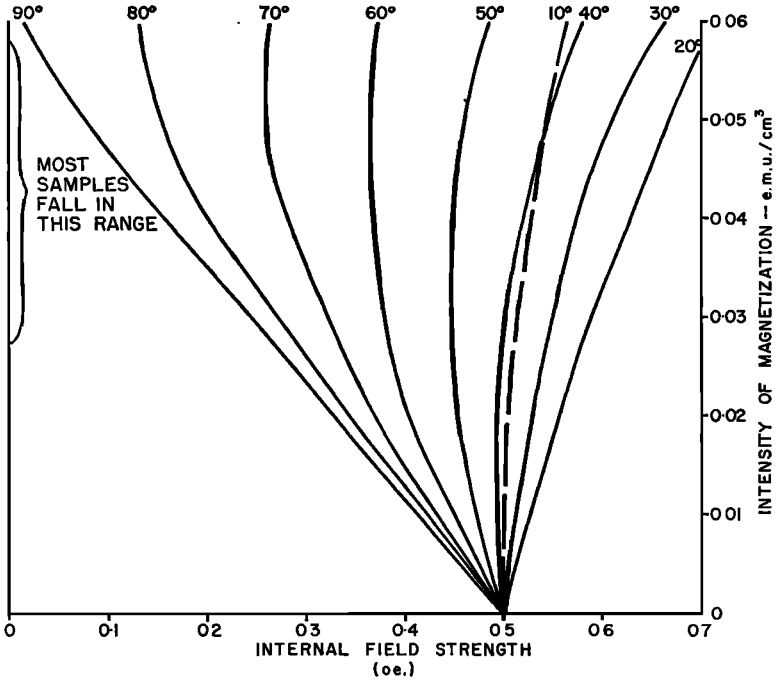


Fig. 10. Variation of internal field strength for various intensities of magnetization and various directions of permanent magnetization (including Lorentz field).

tion of magnetization changed gradually until it approached that of the second applied field. This is illustrated for a typical sample in Figure 9. This experiment established the presence of a stable, low-intensity component of magnetization, having a lower Curie temperature and much greater stability than the main component.

It is postulated that this is the magnetization whose direction is preserved throughout the lifetime of the diabase. Some of the other components may also be preserved, but it seems likely that the softer component will decay more rapidly with time and be more subject to alteration.

The model may also provide an explanation of certain other aspects of the natural magnetization process. If the applied external field is originally normal to the dike, the intensity of the internal field becomes very small (see Fig. 10). *Nagata* [1953] has shown that the intensity of magnetization is proportional to the applied field. As a result, in this case the stable component of magnetization will be very small. This possibility may explain the results obtained for dikes 2 and 7, where no measurable stable component was found. When samples from these dikes were allowed to cool through their Curie temperatures, stable components of magnetization were produced indicating that their magnetic properties are similar to those of the other dikes.

Source of stable component. The rocks were examined for possible sources of the hard component of magnetization. About 30 thin sections and 30 polished sections were studied to obtain some idea of the distribution of the iron oxides. In general, the samples contained 5 to 10 per cent of these minerals. In coarse-grained samples, large grains (about 0.5 mm) of these minerals were found as intimate sheetlike intergrowths of magnetite and ilmenite (see Fig. 11). The Curie point measurements suggest that magnetite is almost always present as a discrete phase. If it were in solid solution with ilmenite, the Curie point would be lowered. The laths of magnetite and ilmenite are probably only a few microns thick, although this dimension is difficult to estimate since most of the polished sections are not cut normal to the intergrowth plane. This conclusion is supported by the fact that the fine-grained edge samples with no in-

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tergrowths have almost identical magnetic properties to the coarse-grained center samples.

Distributed in the coarse-grained portions some finer-grained iron oxides having dimensions of the order of a few microns were also found. These grains may be the low Curie point component. However, it seems unlikely that this is the source, as all the grains in the edge samples have approximately these same dimensions but still have the same magnetic properties. In the study of the thin sections, however, another distribution of opaque mineral believed to be magnetite was found, occurring in the olivine and pyroxene grains. This is a powder distribution of magnetite, apparently formed by reaction of the two host minerals with residual water-rich solutions in the hydrothermal stage of rock formation. The powder magnetite grains are less than a micron and hence close to theoretical single domains in size. *Néel's* [1955] theory of single-domain grains suggests that grains having dimensions of about 0.05 μ will have remanent magnetizations of great stability. It is suggested that this powder magnetite in the olivine and pyroxene (which forms 50 per cent of the rock) may be the site of the stable component of magnetization. In this same theory Néel describes the decrease of Curie temperature with decreasing grain size. The calculation he describes is based on relaxation theory in which thermal fluctuations cause deformation of the individual domains. It can be shown [*Strangway*, 1960], using typical values for magnetite, that for a single domain particle to have a Curie temperature of 500°C and for its magnetic moment to be stable for 10^9 years, its size must be of the order of 0.05 μ.

For these reasons, then, it seems probable that the stable component of magnetization found in this study is due to the powder magnetite formed by reaction within pyroxene and olivine crystals.

Conclusions. It can be stated that a correlation between stable remanent magnetic directions and the strike of diabase dikes has been found. This correlation seems well defined, and a model to explain this anisotropy has been developed. The model depends on the demagnetization due to the shape of a dike and requires that a small portion of the magnetic mineral have a lower Curie temperature and be more stable than the bulk of the magnetic mineral. The presence of such a component has been proved, but it has not been possible to determine which fraction of the iron oxides is the source of this stable magnetization. It is suggested that magnetite formed by the reaction of the pyroxenes and olivines with late-stage hydrothermal iron-rich solutions is the seat of this stable magnetization.

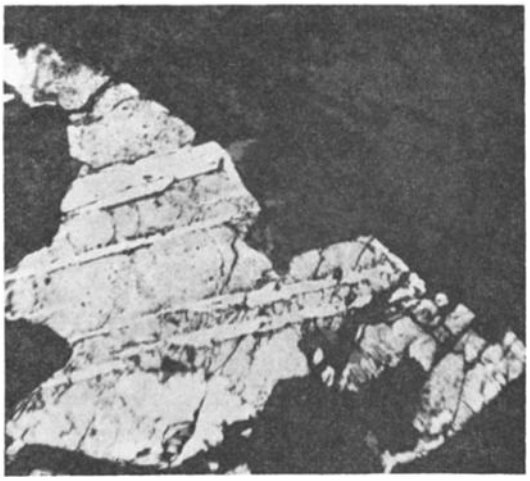


Fig. 11. Photograph in reflected light showing lath structure of iron oxides (width of figure corresponds to 900 microns of specimen).

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